Pact

A 1-to-1 accountability feature for Progression: why it exists, what it does, how to build it.

1. Why this feature

Habit apps lose most users in the first week. People start strong, miss a day, feel guilty and quit. Streaks and XP are private: nobody notices when you stop. A pact adds one trusted person who notices, with no competition, no chat and no strangers.

- **Retention:** a friend who also shows up is a strong reason to return on Day 7 and Day 30.
- **Free growth:** each pact needs an invite, so one user can bring one new user.
- **Low risk:** no free text, nothing to moderate, very little personal data shared.
- **Business value:** real retention data and a social loop raise the value of the app for subscriptions or a sale.

2. What it does

A pact connects exactly two people. Both keep their own goals and missions. The app shows how both are doing and rewards showing up together. The pact works as a team, not a race: there is no winner and no ranking.

Part	What the user sees
Invite	A private link, valid 7 days, one use. Shared through WhatsApp or any app with a ready message.
Pending pact	Before the partner joins, Home shows a "Pact pending" card with a resend button. No dead end.
Agreed minimum	At start both pick a daily minimum of 1, 2 or 3 missions. The higher pick applies to both.
Together days	A weekly count such as **Together 5/7**: days when both people showed up.
Side-by-side card	Each person: missions done out of scheduled (for example "4 of 5"), % done, current streak. Never merged into one score.
Cheer	One preset per day: fire, clap, "do your mission". After a partner misses 2 days, a fourth preset "Missed you" unlocks.
Pact weeks	A week counts when Together days reach 4 or more. The card shows "6 weeks together".
Weekly target	At each rematch both pick a target (Together 5/7 or 6/7). Hit it: a celebration. Miss it: no shame, pick again.
Updated time	Friend data shows "updated 3h ago" and offline states. No fake live status.

3. The user journey

Stage	What happens
Unlock	Invite appears after a personal streak of 5 days or level 3, so users learn the app first.
Ask	The app offers the invite only at a good moment: perfect day, streak 5, 7 or 14, or level-up. At most once per week.
Share	Pre-filled message: "I'm on a 12-day streak. Join my pact?" with the link.
Teaser	Invitee opens the link before any login and sees the inviter's name, streak and this week's %. One button: Accept.
Join	Invitee starts using the app locally, creates the account only at Accept. The invite token survives signup.
First day	Invitee completes a first mission. The Pact card appears on Home for both.
Day 7	Weekly summary, Together days total, rematch and target pick.
Day 30 and on	Pact weeks accumulate. Either person can leave at any time.

If the invite is unopened after 48 hours, the inviter gets one quiet reminder. **If no partner joins,** the card switches to "Beat last week", where the partner is your own previous week. The surface stays useful without a friend.

4. Rules

Showed up on a day means any one of these:

- Done missions are at or above the agreed daily minimum.
- A rest day.
- A streak freeze used.

Sickness, rest and freezes never hurt the partner. One bad day cannot break the shared count.

- **Minimum changes** take effect next Monday and are visible to both people.
- **Time zones:** each person uses their own local date. No shared "crew day".
- **Weeks** start Monday in each person's local date.
- **No combined score.** Together days, done count, percent and streak stay separate.
- **Notifications:** at most one per day, only when the partner finished and you have not. One toggle in Settings.
- **Display name:** up to 20 characters, escaped before display.
- **Pact achievements** reuse the existing achievement list: first pact week, 4 weeks together, 3 rematches. Small XP reward.

5. What friends can see

Shared	Never shared
Display name, date, scheduled count, done count, XP, streak, rest-day flag	Goal names, mission names, sleep, water, custom trackers, email, notes

The privacy policy lists these exact fields. Deleting an account removes every pact row, daily summary and cheer.

6. Backend design (Supabase)

Pact data is separate from full-state sync. Each person writes only their own row for each day, so there is nothing to merge and no conflict.

Table	Purpose
profiles	user_id, display_name (max 20 characters)
pacts	id, invite_token (random 128-bit, unique), created_by, status, expires_at, week_target
pact_members	pact_id, user_id, joined_at, left_at, daily_min. Unique per pact and user. Max 2 current members.
daily_stats	user_id, date, week_start, scheduled, done, xp, rested, streak, updated_at. Key: user_id and date.
cheers	from_user, to_user, date, kind. Key: from_user and date, so one cheer per day.

- **Clients cannot insert or update** pact_members or daily_stats directly. All writes go through server functions.
- **redeem_invite(token):** checks expiry, status, member count, then adds the member.
- **submit_day(...):** idempotent upsert of the caller's own row. The app queues days in local storage and retries with backoff when offline.
- **leave_pact(pact_id):** sets left_at at once. Offline, the button is disabled with the message "Connect to leave". Leaving is never faked.
- Reading a friend's stats needs both people to be current members of the same pact (left_at empty).
- Indexes: daily_stats (user_id, date), pact_members (user_id) and (pact_id) filtered on left_at is null, pacts (invite_token).

Core SQL

```
create function public.is_pact_peer(other uuid)
returns boolean language sql security definer stable
set search_path = public as $$
  select exists (
    select 1 from pact_members a
    join pact_members b on a.pact_id = b.pact_id
    where a.user_id = auth.uid() and b.user_id = other
      and a.left_at is null and b.left_at is null
  );
$$;

alter table daily_stats enable row level security;
create policy read_stats on daily_stats for select
using (user_id = auth.uid() or public.is_pact_peer(user_id));
-- no insert/update policy for clients

create function public.submit_day(p_date date, p_scheduled int,
  p_done int, p_xp int, p_rested bool, p_streak int)
returns void language plpgsql security definer
set search_path = public as $$
begin
  if auth.uid() is null then raise exception 'auth'; end if;
  if p_done > p_scheduled
     or p_date > (now() at time zone 'utc')::date + 1
     or p_xp > p_scheduled * 120 + 700
  then raise exception 'invalid'; end if;
  insert into daily_stats(user_id, date, week_start, scheduled,
    done, xp, rested, streak, updated_at)
  values (auth.uid(), p_date, date_trunc('week', p_date)::date,
    p_scheduled, p_done, p_xp, p_rested, p_streak, now())
  on conflict (user_id, date) do update set
    scheduled = excluded.scheduled, done = excluded.done,
    xp = excluded.xp, rested = excluded.rested,
    streak = excluded.streak, updated_at = now();
end $$;
```

```
create function public.leave_pact(p_pact uuid)
returns void language sql security definer
set search_path = public as $$
  update pact_members set left_at = now()
  where pact_id = p_pact and user_id = auth.uid()
    and left_at is null;
$$;
```

Security-definer functions run as table owner. Keep each one short, fix search_path, and validate every input. The helper is_pact_peer is read-only to avoid recursive policy checks.

7. Security tests before launch

- Invite token guessing, replay after use, and use after expiry.
- A third person joining a full pact.
- Unauthorized removal of another member.
- Leaving, then reading old stats: must return zero rows.
- A stranger or a different pact reading any stats: zero rows.
- Forged stats: done above scheduled, impossible XP, future dates, odd time zones.
- Two saves at the same moment for the same day.
- Cheer spam (blocked by one-per-day key).
- Direct client insert into pact_members or daily_stats: must fail.
- Deleted account leaves no rows.

8. Prerequisites

The feature handles other people's data, so these come first. Full sync hardening is not required, because pact data uses its own one-row-per-day path.

- Sign-out button
- Delete account including cloud rows
- Row-level security automated test
- Auth settings review (rate limits, redirect URLs, token expiry)

9. How to measure it

Events: invite_created, invite_opened, invite_accepted, signup_from_invite, pact_joined, first_shared_day, cheer_sent, pact_week_done, target_hit, rematch, pact_left. Properties: streak bucket and together days only. No names, no goal text.

Stage	Result	Decision
Invite funnel	Open rate under 40% of invites	Fix message and timing
	Accept rate under 25% of opens	Fix teaser page
Closed beta (15 to 20 users, direction only)	0 pacts from 15 invites	Remove feature
	Fewer than 2 pacts	Fix invite flow
	3 or more pacts and 1 or more rematch	Keep going
Retention of pacts	Week-4 survival under 40% of formed pacts	Rethink reinforcement

Stage	Result	Decision
After public launch (about 100 pacts)	Pact users D7 at least 10 points above matched solo users	Keep
	5 to 9 points above	Iterate
	Under 5 points	Remove

Compare pact users with solo users at a similar streak level. Motivated users join pacts anyway, so a plain comparison would overstate the effect. Small beta numbers show direction only, not proof.

10. Scope and timing

Build in first version	Later
Unlock gate, invite link, pending card, teaser page, agreed minimum, Pact card, Together days, pact weeks, weekly target, one cheer a day, submit_day, leave, delete cascade, rematch, solo "Beat last week"	Groups of 3 to 5, share images, friend-activity feed, rankings, leaderboards

- **Placement:** post-launch (P2). Start after the public launch decision, once sign-out, delete and security tests are done.
- **Pricing:** the 1-to-1 pact is free, as a growth loop. Larger groups may become a Pro feature.
- **Biggest product risk:** users do not send invites.
- **Biggest technical risk:** a security-rule mistake leaking data.
- **Honest limit:** interest may fade after month two. Week-4 and week-8 pact survival is the signal to watch.